41.3: Organs specialized for sequential stages of food processing form the mammalian digestive system

In mammals, a number of accessory glands support food processing by secreting digestive juices through ducts into the alimentary canal. There are three pairs of salivary glands, as well as three individual glands: the pancreas, the liver, and the gallbladder. To explore the coordinated function of the accessory glands and alimentary canal, we'll consider the steps in food processing as a meal travels along the canal in a human.

The Oral Cavity, Pharynx, and Esophagus

As soon as food enters your mouth, or **oral cavity**, food processing begins **(Figure 41.8)**. Teeth with specialized shapes cut, mash, and grind, breaking the food into smaller pieces. This mechanical breakdown not only increases the surface area available for chemical breakdown but also facilitates swallowing. Meanwhile, the anticipation or arrival of food in the oral cavity triggers the release of saliva by the **salivary glands**.

Figure 41.8

The human digestive system.

The human digestive system. Structures are labeled and accessory glands are highlighted. The esophagus is a long, straight tube, while the intestines are heavily folded.

The human digestive system, described in order of the appearance of structures beginning with the mouth: mouth (with salivary glands highlighted as accessory glands); esophagus; stomach; small intestine (with accessory glands gall bladder, liver and pancreas highlighted); large intestine; rectum; anus.

After food is chewed and swallowed, it takes 5–10 seconds for it to pass down the esophagus and into the stomach, where it is stored for 2–6 hours during the first stages of processing. Complete digestion and nutrient absorption occur in the small intestine over a period of 5–6 hours. Processing is completed in the large intestine, and undigested material is expelled through the anus as feces.

Animation

Overview of the Human Digestive System

image

image

Saliva is a complex mixture of materials with a number of vital functions. One major component is **mucus**, a viscous mixture of water, salts, cells, and slippery glycoproteins (carbohydrate-protein complexes). Mucus lubricates food for easier swallowing, protects the gums against abrasion, and facilitates taste and smell. Saliva also contains buffers, which help prevent tooth decay by neutralizing acid, and antimicrobial agents (such as lysozyme; see Figure 5.16), which protect against bacteria that enter the mouth with food.

The most abundant enzyme in saliva is **amylase**, which breaks down starch (a glucose polymer from plants) and glycogen (a glucose polymer from animals). While very little digestion occurs in the mouth, the action of chewing mixes amylase throughout the food where it can continue its digestive actions farther along the alimentary canal.

The tongue also has important roles in food processing. Taste buds on the tongue help evaluate ingested material, selecting which foods should be processed further and then enabling their passage. (See Concept 50.4 for a discussion of the sense of taste.) Once chewing begins, tongue movements manipulate the mixture of saliva and food, helping shape it into a ball called a *bolus***(Figure 41.9)**. During swallowing, the tongue provides further assistance, pushing the bolus to the back of the oral cavity and into the pharynx.

Figure 41.9

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Intersection of the human airway and digestive tract.
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The intersection of the human airway and digestive tract with a. trachea open and b. esophagus open.

In humans, the pharynx connects to the trachea and the esophagus. ***(a)*** *At most times, a contracted sphincter seals off the esophagus while the trachea remains open.* ***(b)*** *When a food bolus arrives at the pharynx, the swallowing reflex is triggered. Movement of the larynx, the upper part of the airway, tips a flap of tissue called the epiglottis down, preventing food from entering the trachea. At the same time, the esophageal sphincter relaxes, allowing the bolus to pass into the esophagus. The trachea then reopens, and peristaltic contractions of the esophagus move the bolus to the stomach.*

The intersection of the human airway and digestive tract with, a, trachea open and, b, esophagus open. Order of structures is: mouth with tongue and epiglottis (flap) at back, followed by the pharynx, which branches to, 1, the glottis and trachea; or 2, the esophageal sphincter and then the esophagus.

Part a. Trachea, leading to lungs, is open. Food bolus is in mouth. Epiglottitis is up. The esophageal sphincter at the bottom of the pharynx is closed, cutting off the route to the esophagus.

Part b. Esophagus is open. Epiglottis is down and glottis is up and closed. Food bolus is entering pharynx. Esophageal sphincter is relaxed, allowing food to pass into esophagus.

VISUAL SKILLS *If you laugh while drinking water, the liquid may be ejected from your nostrils. Use this diagram to explain why this happens, taking into account that laughing involves exhaling.*

*For suggested answer, see*Appendix A.**

Each bolus of food is received by the **pharynx**, or throat region, which leads to two passageways: the esophagus and the trachea. The **esophagus** is a muscular tube that connects to the stomach; the trachea (windpipe) leads to the lungs. Swallowing must therefore be carefully choreographed to keep food and liquids from entering the trachea and causing choking, a blockage of the trachea. The resulting lack of airflow into the lungs can be fatal if the material is not dislodged by vigorous coughing, a series of back slaps, or a forced upward thrust of the diaphragm (the Heimlich maneuver).

Within the esophagus, food is pushed along by **peristalsis**, alternating waves of smooth muscle contraction and relaxation. Upon reaching the end of the esophagus, the bolus encounters a **sphincter**, a ringlike valve of muscle **(Figure 41.10)**. Acting like a drawstring, the sphincter regulates passage of the ingested food into the next compartment, the stomach.

Figure 41.10

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The stomach and its secretions.
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1, Interior surface of the stomach, with cell types and functions detailed, and 2, outline of the production of gastric juice.

Parts are as follows.

1. Interior surface of stomach. The interior surface of the stomach wall is highly folded and dotted with pits leading into tubular gastric glands. Stomach wall is lined with simple columnar epithelium cells. Gastric glands. The gastric glands have three types of cells that secrete different components of the gastric juice: mucous cells, chief cells, and parietal cells. Mucous cells secrete mucous, which lubricates and protects the cells lining the stomach. Chief cells secrete pepsinogen, an inactive form of the digestive enzyme pepsin. Parietal cells produce the components of hydrochloric acid (H C l).
2. The production of gastric juice. A close look at interior stomach surface cells reveals a single layer of cells that consists of chief cells and larger parietal cells, here curved into a U shape with the lumen of the stomach at the center. 1. Pepsinogen and H C l are introduced into the lumen of the

stomach when parietal cells secrete H plus and C l minus ions and chief cells secrete pepsinogen. 2. H C l converts pepsinogen to pepsin (active enzyme). 3. Pepsin then activates more pepsinogen, starting a chain reaction. Pepsin begins the chemical digestion of proteins.

Figure 41.10 Walkthrough

The stomach and its secretions

Watch an explanation of this figure in the Figure 41.10 Walkthrough: The stomach and its secretions.

image

image

Digestion in the Stomach

The **stomach**, which is located just below the diaphragm, has two major roles in digestion. The first is storage. With a very elastic wall and accordion-like folds, the stomach can stretch to accommodate about 2 L of food and fluid. The second major function is to process food into a liquid suspension. The stomach secretes a digestive fluid called **gastric juice** and mixes it with the food through a churning action. This mixture of ingested food and gastric juice is called **chyme**.

Chemical Digestion in the Stomach

Two components of gastric juice help liquefy food in the stomach. First, hydrochloric acid (HCl) disrupts the extracellular matrix that binds cells together in meat and plant material. The concentration of HCl is so high that the pH of gastric juice is about 2, acidic enough to dissolve iron nails (and to kill most bacteria). This low pH denatures (unfolds) proteins in food, increasing exposure of their peptide bonds. The exposed bonds are then attacked by the second component of gastric juice—a **protease**, or protein-digesting enzyme, called **pepsin**. Unlike most enzymes, pepsin is adapted to work best in a very acidic environment. By breaking peptide bonds, it cleaves proteins into smaller polypeptides and further exposes the contents of ingested tissues.

Two types of cells in the gastric glands of the stomach produce the components of gastric juice (see Figure 41.10). *Parietal cells* use an ATP-driven pump to expel hydrogen ions into the lumen. At the same time, chloride ions diffuse into the lumen through specific membrane channels of the parietal cells. It is therefore only within the lumen that hydrogen and chloride ions combine to form HCl. Meanwhile, *chief cells* release pepsin into the lumen in an inactive form called **pepsinogen**. HCl converts pepsinogen to active pepsin by clipping off a small portion of the molecule and exposing its active site. Through these processes, both HCl

and pepsin form in the lumen (cavity) of the stomach, not within the cells of the gastric glands. As a result, the parietal and chief cells produce gastric juice but are not digested from within by its components.

After hydrochloric acid converts a small amount of pepsinogen to pepsin, pepsin itself helps activate the remaining pepsinogen. Pepsin, like HCl, can clip pepsinogen to expose the enzyme's active site. This generates more pepsin, which activates more pepsinogen. This series of events is an example of positive feedback (see Concept 40.2).

Why don't HCl and pepsin eat through the lining of the stomach? For one thing, mucus secreted by cells in gastric glands protects against self-digestion (see Figure 41.10). In addition, cell division adds a new epithelial layer every three days on average, replacing cells before the lining is fully eroded by digestive juices.

Stomach Dynamics

The breakdown of food by gastric juices is enhanced by muscular activity of the stomach. The coordinated series of muscle contractions and relaxations that we call "churning" mixes the stomach contents about every 20 seconds. Churning facilitates the action of HCl and pepsin by bringing all of the food into contact with the gastric juices secreted by the lining of the stomach. As a result, what began as a recently swallowed meal becomes the acidic, nutrient-rich broth known as chyme.

Contractions of stomach muscles also help move material through the alimentary canal. In particular, peristaltic contractions typically move the contents of the stomach into the small intestine within 2–6 hours after a meal. The sphincter located where the stomach opens to the small intestine helps regulate passage into the small intestine, allowing only one squirt of chyme to enter at a time.

Occasionally, the sphincter at the top of the stomach allows a movement, or flux, of chyme from the stomach back into the lower end of the esophagus. The painful irritation of the esophagus that results from this process of acid reflux is commonly called "heartburn."

Digestion in the Small Intestine

Although there is some chemical digestion in the oral cavity and stomach, most enzymatic hydrolysis of macromolecules from food occurs in the small intestine **(Figure 41.11)**. This organ's name refers to its small diameter compared to the large intestine, not to its length. The **small intestine** is in fact the alimentary canal's longest compartment—over 6m 20feet long in humans. The first 25 cm 10inches or so of the small intestine forms the **duodenum**. It is here that chyme from the stomach mixes with digestive juices from the pancreas, liver, and gallbladder, as well as from gland cells of the intestinal wall itself. As you

will read in Concept 41.5, hormones released by the stomach and duodenum control the digestive secretions into the alimentary canal.

Figure 41.11

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Chemical digestion in the human digestive system.
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A table provides the sites for the digestion of carbohydrates, proteins, nucleic acids, and fats.

Chemical breakdown of nutrients occurs predominantly in the small intestine, making use of digestive enzymes specific to each class of nutrients.

Nutrient Class	Structure	Substrates	Enzymes	Products
Carbohydrate digestion	Oral cavity, pharynx, esophagus	Polysaccharides (starch, glycogen)	Salivary amylase	Smaller poly-saccharides, maltose
Carbohydrate Digestion	Small intestine (enzymes from pancreas)	Smaller polysaccharides	Pancreatic amylases	Disaccharides
Carbohydrate Digestion	Small intestine (enzymes from intestinal epithelium)	Disaccharides	Disaccharidases	Monosaccharides
Protein Digestion	Stomach	Proteins	Pepsin	Small polypeptides
Protein Digestion	Small intestine (enzymes from pancreas)	Small polypeptides	Pancreatic trypsin and chymotrypsin (These protein-digesting enzymes, or proteases, cleave bonds adjacent to certain amino acids).	Smaller polypeptides
Protein Digestion	Small intestine (enzymes from	Small peptides	Pancreatic carboxypeptidase	Smaller peptides, amino acids

	intestinal epithelium)			
Protein Digestion	Small intestine (enzymes from intestinal epithelium)	Small peptides	Dipeptidases, carboxypeptidase, aminopeptidase. (These proteases each split off one amino acid at a time from a dipeptide or polypeptide).	amino acids
Nucleic Acid Digestion	Small intestine (enzymes from pancreas)	DNA, RNA	Pancreatic nucleases	Nucleotides
Nucleic Acid	Small intestine (enzymes from intestinal epithelium)	Nucleotides	Nucleotidases	Nucleosides
Nucleic Acid	Small intestine (enzymes from intestinal epithelium)	Nucleosides	Nucleosidases and	Nitrogenous bases, sugars, phosphates

Pepsin is resistant to the denaturing effect of low pH *and is thus adapted to the environment of the stomach. Thinking about the small intestine, describe a different adaptation common to the digestive enzymes in that compartment.*

*For suggested answer, see*Appendix A*.*

Animation

Digestive System Function

image

The arrival of chyme in the duodenum triggers release of the hormone secretin, which stimulates the **pancreas** to secrete bicarbonate. Bicarbonate neutralizes the acidity of chyme and acts as a buffer for chemical digestion in the small intestine. The pancreas also secretes numerous digestive enzymes into the small intestine. These include the proteases trypsin and chymotrypsin, which are produced in inactive forms. In a chain reaction similar to that for pepsinogen, they are activated when safely located in the lumen of the duodenum.

The epithelial lining of the duodenum is the source of additional digestive enzymes. Some are secreted into the lumen of the duodenum, whereas others are bound to the surface of epithelial cells. Together with the enzymes from the pancreas, they complete most digestion in the duodenum.

Fats present a particular challenge for digestion. Insoluble in water, they form large globules that cannot be attacked efficiently by digestive enzymes. In humans and other vertebrates, fat digestion is facilitated by bile salts, which act as emulsifiers (detergents) that break apart fat and lipid globules. Bile salts are a major component of **bile**, a secretion of the **liver** that is stored and concentrated in the **gallbladder**.

Bile production is metabolically linked to another vital liver function: the destruction of red blood cells that are no longer fully functional. Pigments released during red blood cell disassembly are incorporated into bile pigments, which are eliminated from the body with the feces. In some liver and blood disorders, bile pigments accumulate in the skin, resulting in a yellowing called jaundice.

Absorption in the Small Intestine

With digestion largely complete, the contents of the duodenum move into the *jejunum* and *ileum*, the remaining regions of the small intestine. There, nutrient absorption occurs across the lining of the intestine **(Figure 41.12)**. Large folds in the lining encircle the intestine and are studded with finger-shaped projections called **villi** (singular, *villus*). Within the villi, each epithelial cell has many microscopic projections, or **microvilli**, that face the intestinal lumen. These epithelial microvilli have a brush-like appearance that is reflected in the name *brush border*. Together, the folds, villi, and microvilli have a surface area of 200–300 m2, roughly the size of a tennis court. This enormous surface area is an evolutionary adaptation that greatly increases the rate of nutrient absorption (see Figure 33.9 for more discussion and examples of maximizing surface area in diverse organisms).

Figure 41.12

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Nutrient absorption in the small intestine.
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Nutrients absorbed by villi in intestinal wall, with a closer look at how nutrients move from the intestinal lumen into epithelial cells and across the basal surface towards the capillaries.

Water-soluble nutrients, such as amino acids and sugars, enter the bloodstream, whereas fats are transported to the lymphatic system.

Long Description: [DO NOT DELETE THIS LINE]

The structure of the small intestine is a heavily folded tube. On the outer surface are muscle layers and blood vessels, including veins carrying blood to the liver. The inner small intestinal wall is lined with brush-like villi arranged in large circular folds; nutrients are absorbed by the villi. A closer look at the villi reveals finger-shaped projections covered with blood capillaries and lacteals that run into larger blood and lymph vessels at the finger bases. Epithelial cells line each finger projection. Nutrients are absorbed into the capillaries. A closer look at the absorption of nutrients in the villi reveals that nutrients are absorbed from the lumen of the small intestine into epithelial cells lining the villi, and the nutrients pass through the epithelial cells and out the basal surface to the capillary.

Tapeworms sometimes infect the human alimentary canal, anchoring themselves to the wall of the small intestine. Based on how digestion is compartmentalized along the mammalian alimentary canal, what digestive functions would you expect these parasites to have?

*For suggested answer, see*Appendix A.**

BioFlix® Animation

Nutrient Transport Across Membranes

image

Depending on the nutrient, transport across the epithelial cells can be passive or active (see Concepts 7.3 and 7.4). The sugar fructose, for example, moves by facilitated diffusion down its concentration gradient from the lumen of the small intestine into the epithelial cells. From there, fructose exits the basal surface and is absorbed into microscopic blood vessels, or capillaries, at the core of each villus. Other nutrients, including amino acids, small peptides, vitamins, and most glucose molecules, enter the epithelial cells of the villus by being pumped against concentration gradients. This active transport allows much more absorption of those nutrients than would be possible with passive diffusion alone.

The capillaries and veins that carry nutrient-rich blood away from the villi converge into the **hepatic portal vein**, a blood vessel that leads directly to the liver. From the liver, blood travels to the heart and then to other tissues and organs. This arrangement serves two major functions. First, it allows the liver to regulate the distribution of nutrients to the rest of the body. Because the liver converts many organic nutrients to different forms for use elsewhere, blood leaving the liver may have a very different nutrient balance than the blood that entered. Second, the arrangement allows the liver to remove toxic substances before they can circulate broadly. The liver is the primary site for detoxifying many organic molecules foreign to the body, such as drugs, and certain metabolic waste products.

Although many nutrients leave the small intestine through the bloodstream and pass through the liver for processing, some products of fat (triglyceride, also known as triacylglycerol) digestion take a different path **(Figure 41.13)**. Hydrolysis of a fat by lipase in the small intestine generates fatty acids and a monoglyceride

(glycerol joined to a fatty acid). These products are absorbed by epithelial cells and recombined into triglycerides. They are then coated with phospholipids, cholesterol, and proteins, forming globules called **chylomicrons** that are too large to enter blood vessels.

Figure 41.13

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Digestion and absorption of fats.
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Digestion and absorption of fats by the small intestine includes the enzymatic hydrolysis of triglycerides in the lumen and the formation of chylomicrons inside the epithelial cells.

Fats, which are insoluble in water, are broken down in the lumen of the small intestine, reassembled in epithelial cells, and then transported in water-soluble globules called chylomicrons. The chylomicrons enter the lymph via narrow vessels called lacteals and are later transferred to the blood in large veins leading to the liver and heart.

Steps are as follows.

1. In the lumen of the small intestine, bile salts break up large fat globules into fat droplets, increasing exposure of triglycerides (fat molecules) on the surface to hydrolysis.
2. During enzymatic hydrolysis, the enzyme lipase breaks the exposed triglycerides down to fatty acids and monoglycerides.
3. After diffusing into epithelial cells, monoglycerides and fatty acids are re-formed into triglycerides. (Some glycerol and fatty acids pass directly into capillaries.)
4. The triglycerides are incorporated into particles called chylomicrons. Phospholipids and proteins on the surface make chylomicrons water-soluble.
5. Chylomicrons leave epithelial cells by exocytosis and enter lacteals, where they are carried away by the lymph and later pass into large veins that lead directly to the heart.

VISUAL SKILLS *Arrows at two locations in this figure indicate movement of materials between the epithelial cell and its surroundings. Identify and circle these arrows. At either of these locations, do the movements represented require an input of energy? Explain.*

*For suggested answer, see*Appendix A.**

HHMI BioInteractive Animation

The Fate of Fat

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In exiting the small intestine, chylomicrons first enter a **lacteal**, a vessel at the core of each villus that has large gaps between adjacent cells (see Figure 41.13, step 5). Lacteals are part of the vertebrate lymphatic system, which is a network of vessels filled with a clear fluid called lymph. Starting at the lacteals, lymph containing the chylomicrons passes into the larger vessels of the lymphatic system and eventually into large veins that return the blood directly to the heart.

In addition to absorbing nutrients, the small intestine recovers water and ions. Each day a person consumes about 2 L of water and secretes another 7 L in digestive juices into the alimentary canal. Typically all but 0.1 L of the water is reabsorbed, mostly in the small intestine. There is no mechanism for active transport of water. Instead, water is reabsorbed by osmosis when sodium and other ions are pumped out of the lumen of the small intestine.

Processing in the Large Intestine

The alimentary canal ends with the **large intestine**, which includes the colon, cecum, and rectum. The small intestine connects to the large intestine at a T-shaped junction **(Figure 41.14)**. One arm of the T is the 1.5-m-long **colon**, which leads to the rectum and anus. The other arm is a pouch called the **cecum**. In animals that eat large amounts of plant material, the cecum has an important role in fermenting ingested material. In humans, the cecum is small and has an **appendix**, a finger-shaped extension that acts as a reservoir for symbiotic microorganisms, which are discussed in Concept 41.4.

Figure 41.14

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Junction of the small and large intestines.
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Junction of small and large intestines with small intestine connecting at point below the ascending portion of colon and above the cecum and appendix.

The colon completes the recovery of water that began in the small intestine. What remain are the **feces**, the wastes of the digestive system, which become increasingly solid as they are moved along the colon by peristalsis. It takes approximately 12–24 hours for material to travel the length of the colon. The undigested material in feces includes cellulose fiber. Although it provides no caloric value (energy) to humans, it helps move food along the alimentary canal.

If the lining of the colon is irritated—by a viral or bacterial infection, for instance—less water than normal may be reabsorbed, resulting in diarrhea. The opposite problem, constipation, occurs when the feces move

along the colon too slowly. Too much water is reabsorbed, and the feces become compacted.

The community of bacteria living on unabsorbed organic material in the human colon contributes about one-third of the dry weight of feces. As by-products of their metabolism, many colon bacteria generate gases, including methane and hydrogen sulfide, the latter of which has an offensive odor. These gases and ingested air are expelled through the anus.

The terminal portion of the large intestine is the **rectum**, where the feces are stored before elimination. Two sphincters separate the rectum and the anus; the inner one is involuntary and the outer one is voluntary. Periodically, strong contractions of the colon create an urge to defecate. Because filling of the stomach triggers a reflex that increases the rate of contractions in the colon, the urge to defecate often follows a meal.

Having followed a meal through the alimentary canal, we'll look next at some adaptations of this general digestive plan in different animals.